TECHNICAL & CODING

100 PYQ-Style Questions — Practice Set 1

Placement & Competitive Exam Practice

Included	100 Distinct PYQ-Style Questions • Practice Set 1
Useful For	TCS NQT • Cognizant • Infosys • Wipro • Accenture • Bosch • Campus Placements
Topics Covered	C • C++ • Java • Python • Data Structures • DBMS & SQL • JavaScript • OS • Networks • OOPs
Extra	Answer Key on Each Page • Practice Set 1 • 1 Page per Language

Structure: 10 Core Programming Languages & Technical Domains. Set 1 contains 100 unique questions. Each language contains exactly 10 authentic MCQs with an Answer Key box at the bottom of the page.

TECHNICAL ASSESSMENT • C PROGRAMMING

Practice Set 1 • Section 1 of 10 • 10 Questions with Answer Key

Q1. What is the return value of sizeof('a') in standard C?

A) 1
B) 2
C) 4 (sizeof int)
D) Compiler Error

Q2. What does malloc() return if the operating system cannot allocate the requested memory?

A) 0
B) -1
C) NULL
D) void

Q3. Which of the following operators cannot be overloaded or redefined in C?

A) * (dereference)
B) sizeof
C) & (address-of)
D) -> (member)

Q4. What is the default storage class for a local variable declared inside a C function?

A) static
B) auto
C) extern
D) register

Q5. Which format specifier is used to print a memory address stored in a pointer in printf?

A) %u
B) %p
C) %x
D) %d

Q6. Which standard library header file contains declarations for malloc(), calloc(), and free()?

A) <stdio.h>
B) <stdlib.h>
C) <string.h>
D) <memory.h>

Q7. In C, what is the value of EOF as defined in <stdio.h>?

A) 0
B) 1
C) -1
D) NULL

Q8. What is a dangling pointer in C programming?

A) A pointer initialized to NULL
B) A pointer pointing to deallocated memory
C) A void pointer
D) An uninitialized pointer

Q9. What does the volatile keyword tell the C compiler regarding a variable?

A) Variable is read-only
B) Value can change unexpectedly; do not cache in registers
C) Variable is thread-safe
D) Store in CPU cache

Q10. Which function is used to dynamically allocate memory and initialize all bytes to zero?

A) malloc()
B) calloc()
C) realloc()
D) alloc()

ANSWER KEY & SOLUTIONS — C PROGRAMMING (SET 1)

Q1: C • Q2: C • Q3: B • Q4: B • Q5: B

Q6: B • Q7: C • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • C++ PROGRAMMING & OOPS

Practice Set 1 • Section 2 of 10 • 10 Questions with Answer Key

Q1. Which C++ mechanism is used to achieve runtime (dynamic) polymorphism?

A) Function Overloading
B) Virtual Functions
C) Template Classes
D) Friend Functions

Q2. What is the minimum size of an empty class object in C++?

A) 0 bytes
B) 1 byte
C) 4 bytes
D) Compiler Error

Q3. Which constructor is automatically invoked when an object is passed by value to a function?

A) Default constructor
B) Parameterized constructor
C) Copy constructor
D) Move constructor

Q4. Which operator among the following cannot be overloaded in C++?

A) :: (Scope resolution)
B) + (Addition)
C) [] (Subscript)
D) -> (Arrow)

Q5. What occurs when deleting a derived class object through a base pointer without a virtual destructor?

A) Normal deletion
B) Undefined behavior & partial memory leak
C) Compiler error
D) Segfault always

Q6. What makes a C++ class an abstract class?

A) Having all private variables
B) Containing at least one pure virtual function (= 0)
C) Having no constructor
D) Multiple inheritance

Q7. Which C++ cast operator is used for safe downcasting in polymorphic class hierarchies?

A) static_cast
B) dynamic_cast
C) reinterpret_cast
D) const_cast

Q8. What does a friend function have special access to inside a C++ class?

A) Only public members
B) Public and protected members
C) Private and protected members
D) Global variables only

Q9. What is the primary idiom behind RAII (Resource Acquisition Is Initialization) in modern C++?

A) Managing resource lifetime via object scope/destructor
B) Rapid Application Integration
C) Inline Assembly
D) Static Array Allocation

Q10. Which smart pointer in C++11 enforces exclusive, non-shareable ownership of a resource?

A) std::shared_ptr
B) std::unique_ptr
C) std::weak_ptr
D) std::auto_ptr

ANSWER KEY & SOLUTIONS — C++ PROGRAMMING & OOPS (SET 1)

Q1: B • Q2: B • Q3: C • Q4: A • Q5: B

Q6: B • Q7: B • Q8: C • Q9: A • Q10: B

TECHNICAL ASSESSMENT • JAVA CORE & OOPS

Practice Set 1 • Section 3 of 10 • 10 Questions with Answer Key

Q1. Why is Java not considered a 100% pure object-oriented programming language?

A) Does not support multiple inheritance
B) Supports primitive data types (int, char, etc.)
C) Runs on a virtual machine
D) Lacks operator overloading

Q2. Which JVM memory area stores local variables and method invocation frames?

A) Heap Memory
B) Stack Memory
C) Method Area
D) Native Method Stack

Q3. What is the output of: String a = 'hi'; String b = new String('hi'); System.out.print(a == b);

A) true
B) false
C) Compilation error
D) NullPointerException

Q4. Since Java 8, which keywords allow method implementations directly inside an interface?

A) static and default
B) public and protected
C) abstract and final
D) native and synchronized

Q5. Which Collection framework class is thread-safe and synchronized by default?

A) ArrayList
B) HashMap
C) Vector
D) HashSet

Q6. What is the effect of declaring a Java class as final?

A) It cannot be instantiated
B) It cannot be subclassed/extended
C) All its methods must be static
D) Its members are immutable

Q7. Which exception is thrown when an application attempts to use null where an object is required?

A) NullPointerException
B) ClassCastException
C) IllegalArgumentException
D) NoSuchMethodError

Q8. What is the mandatory contract between equals() and hashCode() in Java?

A) If equals() is true, hashCodes must be equal
B) If hashCodes are equal, equals() must be true
C) Both must return prime numbers
D) No strict contract

Q9. In Java's try-with-resources statement, what interface must the resource implement?

A) java.io.Serializable
B) java.lang.AutoCloseable
C) java.lang.Cloneable
D) java.lang.Runnable

Q10. What happens if both catch and finally blocks have a return statement?

A) Compiler error
B) The finally block return value overrides the catch return
C) The catch block return wins
D) Runtime exception

ANSWER KEY & SOLUTIONS — JAVA CORE & OOPS (SET 1)

Q1: B • Q2: B • Q3: B • Q4: A • Q5: C

Q6: B • Q7: A • Q8: A • Q9: B • Q10: B

TECHNICAL ASSESSMENT • PYTHON PROGRAMMING

Practice Set 1 • Section 4 of 10 • 10 Questions with Answer Key

Q1. What is the output of type(lambda x: x) in Python 3?

A) <class 'function'>
B) <class 'lambda'>
C) <class 'closure'>
D) <class 'type'>

Q2. Which of the following built-in collection types in Python is immutable?

A) list
B) dict
C) set
D) tuple

Q3. What is the output of [x for x in range(6) if x % 2 == 0] in Python?

A) [0, 2, 4]
B) [2, 4, 6]
C) [1, 3, 5]
D) [0, 2, 4, 6]

Q4. What does *args indicate when placed in a Python function definition?

A) Keyword arguments dict
B) Variable number of non-keyword positional arguments
C) Pointer dereference
D) Mandatory argument

Q5. How does CPython primarily perform memory management and garbage collection?

A) Reference counting with cyclic garbage collector
B) Mark and Sweep only
C) Manual memory deallocation
D) Static heap allocation

Q6. What does bool([]) evaluate to in Python?

A) True
B) False
C) None
D) TypeError

Q7. Which method serves as the object constructor/initializer in Python classes?

A) __construct__()
B) __init__()
C) __new__()
D) __create__()

Q8. What is the output of the string slicing expression 'Python'[::-1]?

A) 'Python'
B) 'nohtyP'
C) 'P'
D) 'n'

Q9. What does the Global Interpreter Lock (GIL) do in CPython?

A) Prevents multi-process execution
B) Restricts execution of Python bytecode to one native thread at a time
C) Encrypts source code
D) Prevents recursion

Q10. Which built-in function pairs elements from two iterables into an iterator of tuples?

A) map()
B) zip()
C) enumerate()
D) chain()

ANSWER KEY & SOLUTIONS — PYTHON PROGRAMMING (SET 1)

Q1: A • Q2: D • Q3: A • Q4: B • Q5: A

Q6: B • Q7: B • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • DATA STRUCTURES & ALGORITHMS

Practice Set 1 • Section 5 of 10 • 10 Questions with Answer Key

Q1. What is the worst-case time complexity of QuickSort?

A) O(N log N)
B) O(N)
C) O(N^2)
D) O(log N)

Q2. Which data structure operates strictly on the Last In First Out (LIFO) principle?

A) Queue
B) Stack
C) Binary Heap
D) Linked List

Q3. What is the minimum number of queues required to implement a Stack?

A) 1
B) 2
C) 3
D) 4

Q4. In a Binary Search Tree (BST), which traversal outputs keys in ascending sorted order?

A) Pre-order
B) In-order
C) Post-order
D) Level-order

Q5. What is the worst-case search time complexity in a self-balancing AVL Tree with N nodes?

A) O(1)
B) O(log N)
C) O(N)
D) O(N log N)

Q6. Which algorithmic paradigm does Dijkstra's single-source shortest path algorithm employ?

A) Dynamic Programming
B) Greedy Algorithm
C) Divide and Conquer
D) Backtracking

Q7. What is the worst-case lookup time in a Hash Table when all keys collide into one bucket?

A) O(1)
B) O(log N)
C) O(N)
D) O(N^2)

Q8. Which algorithm is used to detect cycles in a singly linked list using two pointers?

A) Kadane's Algorithm
B) Floyd's Cycle-Finding Algorithm (Tortoise & Hare)
C) KMP Algorithm
D) Prim's Algorithm

Q9. Which of the following sorting algorithms is stable and guarantees O(N log N) worst-case time?

A) QuickSort
B) MergeSort
C) HeapSort
D) SelectionSort

Q10. What data structure is used by the operating system for Breadth-First Search (BFS) graph traversal?

A) Stack
B) Queue
C) Binary Search Tree
D) Trie

ANSWER KEY & SOLUTIONS — DATA STRUCTURES & ALGORITHMS (SET 1)

Q1: C • Q2: B • Q3: B • Q4: B • Q5: B

Q6: B • Q7: C • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • SQL & DATABASE MANAGEMENT (DBMS)

Practice Set 1 • Section 6 of 10 • 10 Questions with Answer Key

Q1. Which SQL clause is used to filter records after aggregate grouping functions like SUM() or COUNT()?

A) WHERE
B) GROUP BY
C) HAVING
D) ORDER BY

Q2. What do the letters in ACID transaction properties stand for?

A) Atomicity, Consistency, Isolation, Durability
B) Access, Control, Integrity, Data
C) Action, Concurrency, Index, Durability
D) Automatic, Commit, Isolate, Direct

Q3. Which Normal Form removes transitive functional dependencies between non-key attributes?

A) 1NF
B) 2NF
C) 3NF
D) BCNF

Q4. Which JOIN returns all rows from the left table and matched rows from the right table?

A) INNER JOIN
B) LEFT (OUTER) JOIN
C) RIGHT JOIN
D) CROSS JOIN

Q5. What is the key difference between DELETE and TRUNCATE commands in SQL?

A) Truncate removes table structure
B) Delete is DML (row-by-row logged), Truncate is DDL (deallocates pages)
C) Delete cannot be rolled back
D) No difference

Q6. What type of index is most commonly used by relational databases for indexing B-Tree/range lookups?

A) Hash Index
B) B+ Tree Index
C) Bitmap Index
D) Linear Index

Q7. What does a FOREIGN KEY constraint enforce in relational database tables?

A) Entity Integrity
B) Referential Integrity
C) Domain Integrity
D) User-defined Integrity

Q8. Which SQL command is classified as a DDL (Data Definition Language) statement?

A) SELECT
B) UPDATE
C) ALTER
D) INSERT

Q9. What does the SQL function COALESCE(val1, val2, val3) return?

A) Sum of non-null values
B) First non-NULL expression in the argument list
C) Maximum value
D) Boolean true/false

Q10. Which transaction isolation level prevents Dirty Reads but still permits Non-Repeatable Reads?

A) Read Uncommitted
B) Read Committed
C) Repeatable Read
D) Serializable

ANSWER KEY & SOLUTIONS — SQL & DATABASE MANAGEMENT (DBMS) (SET 1)

Q1: C • Q2: A • Q3: C • Q4: B • Q5: B

Q6: B • Q7: B • Q8: C • Q9: B • Q10: B

TECHNICAL ASSESSMENT • JAVASCRIPT & WEB TECHNOLOGIES

Practice Set 1 • Section 7 of 10 • 10 Questions with Answer Key

Q1. What is the return value of typeof NaN in JavaScript?

A) 'number'
B) 'NaN'
C) 'undefined'
D) 'object'

Q2. What is a closure in JavaScript?

A) A function that terminates a loop
B) A function bundled with references to its surrounding lexical environment
C) A JSON serializer
D) An anonymous lambda

Q3. Which ES6 keyword declares a block-scoped variable that cannot be reassigned?

A) var
B) let
C) const
D) static

Q4. What is the result of 0.1 + 0.2 === 0.3 in JavaScript?

A) true
B) false (floating-point precision produces 0.30000000000000004)
C) NaN
D) TypeError

Q5. What does event.preventDefault() do when called inside an event listener?

A) Stops event propagation/bubbling
B) Prevents the browser's default action for the event
C) Deletes the event handler
D) Refreshes DOM

Q6. Which method converts a valid JSON string into a native JavaScript object?

A) JSON.stringify()
B) JSON.parse()
C) JSON.toObject()
D) JSON.decode()

Q7. What is the core role of the Event Loop in JavaScript's single-threaded runtime?

A) Compiling JS bytecode
B) Pushing completed async tasks from callback queue onto call stack when stack is empty
C) Garbage collection
D) CSS rendering

Q8. What is the evaluated output of [] + [] in JavaScript?

A) []
B) '' (empty string)
C) undefined
D) NaN

Q9. What does a JavaScript Promise object represent?

A) A synchronous blocking call
B) The eventual completion or failure of an asynchronous operation and its resulting value
C) A multi-threaded worker
D) A network socket

Q10. What is the difference between == and === in JavaScript?

A) == checks memory address, === checks value
B) == performs type coercion before comparison; === checks both value and type without coercion
C) Identical
D) === is deprecated

ANSWER KEY & SOLUTIONS — JAVASCRIPT & WEB TECHNOLOGIES (SET 1)

Q1: A • Q2: B • Q3: C • Q4: B • Q5: B

Q6: B • Q7: B • Q8: B • Q9: B • Q10: B

TECHNICAL ASSESSMENT • OPERATING SYSTEMS & LINUX

Practice Set 1 • Section 8 of 10 • 10 Questions with Answer Key

Q1. Which CPU scheduling algorithm can lead to indefinite starvation of low-priority processes?

A) First-Come, First-Served (FCFS)
B) Round Robin (RR)
C) Priority Scheduling
D) Shortest Job First with Aging

Q2. What is the primary objective of Virtual Memory in modern computer architectures?

A) Increase CPU clock frequency
B) Allow execution of processes larger than physical RAM via paging
C) Encrypt RAM contents
D) Enhance L1 cache size

Q3. What is Thrashing in an operating system memory subsystem?

A) High CPU temperature shutdown
B) High page-fault rate where system spends more time swapping pages than executing instructions
C) Hard disk head collision
D) Buffer overflow crash

Q4. Which Linux terminal command is used to modify read, write, and execute permissions of a file?

A) chown
B) chmod
C) chgrp
D) touch

Q5. Which of the following is NOT one of Coffman's four conditions for Deadlock?

A) Mutual Exclusion
B) Hold and Wait
C) Preemption allowed
D) Circular Wait

Q6. What is a Semaphore in process synchronization?

A) A hardware timer
B) An integer variable accessed via wait() (P) and signal() (V) atomic operations
C) A shell script variable
D) A network socket

Q7. Which Unix/Linux system call creates an exact duplicate child process of the calling process?

A) exec()
B) fork()
C) spawn()
D) pthread_create()

Q8. What is the Process ID (PID) of the systemd / init process in Linux?

A) 0
B) 1
C) -1
D) 100

Q9. Which Linux utility displays dynamic real-time information about CPU, memory, and running tasks?

A) top
B) cat
C) grep
D) sed

Q10. How does a Thread differ from a Process in terms of memory resources?

A) Threads do not share address space
B) Threads of the same process share code, data, and open files, but have independent stacks and registers
C) Threads have isolated heaps
D) No difference

ANSWER KEY & SOLUTIONS — OPERATING SYSTEMS & LINUX (SET 1)

Q1: C • Q2: B • Q3: B • Q4: B • Q5: C

Q6: B • Q7: B • Q8: B • Q9: A • Q10: B

TECHNICAL ASSESSMENT • COMPUTER NETWORKS

Practice Set 1 • Section 9 of 10 • 10 Questions with Answer Key

Q1. Which OSI layer provides reliable host-to-host data transport, flow control, and error recovery?

A) Network Layer
B) Transport Layer
C) Data Link Layer
D) Application Layer

Q2. Which protocol resolves a human-readable domain name (e.g. google.com) into an IP address?

A) DHCP
B) DNS
C) ARP
D) ICMP

Q3. What is the standard default port number used by HTTPS (HTTP Secure) protocol?

A) 80
B) 21
C) 443
D) 8080

Q4. What are the key characteristics distinguishing TCP from UDP?

A) UDP guarantees in-order delivery
B) TCP is connection-oriented and reliable; UDP is connectionless, lightweight, and low-latency
C) TCP lacks congestion control
D) Both are connectionless

Q5. Which protocol resolves a target IP address to its corresponding physical MAC address on a LAN?

A) ARP (Address Resolution Protocol)
B) RARP
C) DHCP
D) NAT

Q6. What is the total bit length of an IPv4 address compared to an IPv6 address?

A) IPv4: 32 bits, IPv6: 128 bits
B) IPv4: 64 bits, IPv6: 128 bits
C) IPv4: 16 bits, IPv6: 64 bits
D) IPv4: 32 bits, IPv6: 64 bits

Q7. Which HTTP status code signifies '401 Unauthorized' client access?

A) 200
B) 401
C) 403
D) 404

Q8. What sequence of packets completes the TCP Three-Way Handshake to establish a connection?

A) ACK -> SYN -> SYN-ACK
B) SYN -> SYN-ACK -> ACK
C) FIN -> ACK -> FIN-ACK
D) PING -> PONG -> ACK

Q9. In CIDR notation, how many usable host IP addresses are available in a /24 subnet?

A) 256
B) 254 (256 minus network and broadcast IDs)
C) 255
D) 128

Q10. Which protocol is utilized by the ping utility to verify network reachability?

A) TCP
B) UDP
C) ICMP (Internet Control Message Protocol)
D) SNMP

ANSWER KEY & SOLUTIONS — COMPUTER NETWORKS (SET 1)

Q1: B • Q2: B • Q3: C • Q4: B • Q5: A

Q6: A • Q7: B • Q8: B • Q9: B • Q10: C

TECHNICAL ASSESSMENT • OBJECT-ORIENTED PROGRAMMING (OOPS)

Practice Set 1 • Section 10 of 10 • 10 Questions with Answer Key

Q1. Which core OOP principle wraps data attributes and methods into a single unit to prevent unauthorized access?

A) Inheritance
B) Encapsulation
C) Polymorphism
D) Abstraction

Q2. What is method overriding in object-oriented programming?

A) Defining multiple methods with same name and different parameters in the same class
B) A subclass providing its own specific implementation of a method declared in its superclass
C) Overloading operators
D) Writing private helper methods

Q3. Which relationship represents Composition (HAS-A relationship) in OOP design?

A) Car IS-A Vehicle
B) Car HAS-A Engine
C) Dog IS-A Animal
D) Teacher IS-A Person

Q4. What does the 'L' stand for in the industry-standard SOLID design principles?

A) Loose Coupling Principle
B) Liskov Substitution Principle
C) Law of Demeter
D) Linear Inheritance Principle

Q5. Which creational design pattern guarantees that a class has only one instance and provides global access?

A) Factory Pattern
B) Singleton Pattern
C) Observer Pattern
D) Prototype Pattern

Q6. What is the primary distinction between an Abstract Class and an Interface in Java/C++?

A) Interfaces can have state variables
B) A class can implement multiple interfaces but inherit only one base class (in Java/C#)
C) Abstract classes cannot have concrete methods
D) No distinction

Q7. What is the Diamond Problem in multiple inheritance?

A) Memory leak from recursive calls
B) Ambiguity when a class inherits from two parent classes that both derive from a common grandparent
C) Destructor failure
D) Stack overflow

Q8. Which OOP principle focuses on representing only essential features while hiding complex internal details?

A) Abstraction
B) Encapsulation
C) Polymorphism
D) Coupling

Q9. What is dynamic (late) method binding in OOP?

A) Method code address is linked at compile time
B) Method call is resolved at runtime based on the actual object's type
C) Static DLL linking
D) Memory allocation during startup

Q10. Can a constructor in an object-oriented class be declared as virtual?

A) Yes, in C++ and Java
B) No, constructors cannot be virtual because object memory/vtable is not yet constructed
C) Only in abstract classes
D) Only if private

ANSWER KEY & SOLUTIONS — OBJECT-ORIENTED PROGRAMMING (OOPS) (SET 1)

Q1: B • Q2: B • Q3: B • Q4: B • Q5: B

Q6: B • Q7: B • Q8: A • Q9: B • Q10: B